

Table 6.2 **Nutrients and Food**

Macronutrients				
	<i>Male</i>	<i>Female</i>	<i>Child</i>	<i>Examples of Food</i>
Water	3.7 L/day	2.7 L/day	1.7 L/day	
Carbohydrates	130 g/day	130 g/day	130 g/day	Bread, Beans, Potato Rice
Protein	56 g/day	46 g/day	19 g/day	Cheese, Milk, Fish, Soya bean
Fiber	38 g/day	25 g/day	25 g/day	Peas, Wheat, Rice
Fat	20–35% of calories		25–35% of calories	Oily fish, Peanuts, Animal fat

Source: Food and Nutrition Board, National Academies, 2004.

states that there is only one factor driving all asset returns, which is the market return in excess of T-bills. All assets have different exposures to the market factor and the greater the exposure, the higher the risk premium. The market is an example of a tradeable, investment factor. Other examples include interest rates, value-growth investing, low volatility investing, and momentum portfolios. Factors can also be fundamental macro factors, like inflation and economic growth. Assets have different payoffs during high or low inflation periods or during economic recessions and expansions. We leave a complete exposition of the various types of factors to the next chapter. In this chapter, we describe the underlying theory of factor risk.

There are three similarities between food and assets:

1. Factors matter, not assets.

If an individual could obtain boring, tasteless nutrients made in a laboratory, she would comfortably meet her nutrient requirements and lead a healthy life. (She would, however, deprive herself of gastronomic enjoyment.) The factors behind the assets matter, not the assets themselves. Investing right requires looking through asset class labels to understand the factor content, just as eating right requires looking through food labels to understand the nutrient content.

2. Assets are bundles of factors.

Foods contain various combinations of nutrients. Certain foods are nutrients themselves—like water—or are close to containing only one type of nutrient, as in the case of rice for carbohydrates. But generally foods contain many nutrients. Similarly, some asset classes can be considered factors themselves—like equities and government fixed income securities—while